Lost Wages and House Prices: The Oil Bust Where Norway's Offshore Workers Lived

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Abstract

The studies of the 2014 oil price bust measure the shock where the oil industry is located, so the workers and the firms of a region are hit together. I measure it where the workers lived. Norwegian offshore workers fly to the platforms from five heliports and spend two weeks offshore and four at home, so nearly half of them live outside the two oil counties, in municipalities with almost no oil firms. The bust reached these places mostly through the wages of their residents. Using public municipal data from Statistics Norway, I compare the 375 municipalities without an onshore oil economy by the share of their employed residents who worked offshore in 2003. Per percentage point of that share, the average income from employment fell by 1.9 percent between 2013 and 2017, employment of residents by 1.4 percent and the average sale price of dwellings by 7 percent. The dwelling stock and the population did not move. The effects sit on the region where offshore workers live, and the oil firms, the coastal industries and the equipment suppliers left in the sample do not explain them. The loss of income was the men's, and for each offshore job lost the equipment suppliers of the industry lost about one more and the rest of the local economy about two, imprecisely estimated. Lost wages move house prices in places with almost no oil firms.

JEL: R31, R23, J23, Q33 **Keywords:** house prices, local labor markets, oil price shock, offshore workers, income shocks, Norway

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1 Introduction

The 2014 oil price bust has been studied where the oil industry is located, so a region's workers and its firms are hit together. This paper measures it where the workers lived, in municipalities with almost no oil firms, and asks what lost wages do to a local economy. The oil price fell by more than 70 percent between June 2014 and January 2016, after rising from $29 a barrel in 2003 to $109 in 2013. The fall reflected the current and expected increase in oil production and a weakening global demand (Baumeister and Kilian, 2016). The Norwegian oil industry was severely affected by the fall. Between 2013 and 2016, investment in oil and gas extraction and pipelines fell by 25 percent in current prices, and investment in exploration by more than 40 percent. Also, the number of continental shelf workers declined by 26 percent.

Stavanger, home to the headquarters of the largest operator on the continental shelf, Statoil (Equinor since 2018), and areas close to it diverged from the rest of Norway in income, employment and house prices. Between 2013 and 2016, registered unemployment rose by 2.5 percentage points in Rogaland, the county around Stavanger, and barely changed in the rest of Norway. The average sale price of dwellings rose by about 1 percent in Rogaland and by about 21 percent in the rest of Norway.

The studies of the bust follow this contrast. Böjeryd (2025) follows homeowners in the Stavanger region, where the bust lowered earnings and house prices. Aastveit et al. (2025) compare government workers in oil and non-oil regions to measure how spending responded to the fall in house prices. Juelsrud and Wold (2025) compare oil engineers with other skilled workers in the oil region, and Schøne and Strøm (2021) compare couples in labour markets with many and few oil jobs.

The studies of resource booms and busts elsewhere measure them where the industry sits as well. Oil and gas booms raise local wages and incomes (Allcott and Keniston, 2018; Feyrer et al., 2017), house prices rise with them (Bartik et al., 2019), and after a bust incomes fall below the path they would have followed (Jacobsen and Parker, 2016). Each tradable job carries about 1.6 local service jobs with it (Moretti, 2010), and when household wealth falls the jobs that go are the ones that serve local demand (Mian and Sufi, 2014). Notowidigdo (2020) finds that an adverse local shock pushes fewer people out than a favourable one draws in, and estimates how far house prices and transfers carry the loss for those who stay. In all of these places the workers, the firms and the landowners are hit together. The offshore rotation gives a place where only the workers are hit, so the fall in prices can be read as the response to lost pay and the multiplier as the demand of offshore households alone.

The places outside the oil regions were not all alike. Even though few of their local businesses were in the oil industry, some of their residents worked on the continental shelf, in extraction and in the drilling and service jobs around it. In 2003, 6,960 of the 14,912 Norwegian residents whose workplace was the shelf, 47 percent, lived outside the two largest oil counties, Rogaland and Hordaland. They were also spread unevenly (Table 1). In the 375 municipalities without an onshore oil economy, the share of employed residents working offshore was zero in 40 municipalities, 0.28 percent at the median and more than 3 percent in Tysnes, Fedje, Kvinesdal and Bamble. Over the

bust, from 2013 to 2017, the average sale price of dwellings rose by 38 percent in the half of these municipalities with the fewest offshore workers and by 20 percent in the tenth with the most. The average income from employment rose by 10 percent and 6 percent. Compared with Stavanger all of these places did well but compared with one another, those with more offshore workers did worse.

Table 1: The ten municipalities with the most offshore work, 2003

Municipality	Economic region	Employed residents	Working offshore	Share (%)
Tysnes	Sunnhordland	1,271	54	4.25
Fedje	Bergen	282	11	3.90
Kvinesdal	Flekkefjord	2,617	86	3.29
Bamble	Skien/Porsgrunn	6,638	201	3.03
Farsund	Lyngdal/Farsund	4,391	123	2.80
Bømlo	Sunnhordland	5,051	137	2.71
Lyngdal	Lyngdal/Farsund	3,372	83	2.46
Vega	Brønnøysund	629	15	2.38
Tvedestrand	Arendal	2,614	62	2.37
Sauda	Haugesund	2,290	54	2.36
All 375: p10 0.00, p25 0.10, median 0.28, p75 0.75, p90 1.33				
40 at zero; population-weighted mean 0.41				

Note: Residents whose workplace is the continental shelf, fourth quarter of 2003, as a percentage of employed residents (Statistics Norway tables 03321 and 03311), in the 375 municipalities where at most 0.5 percent of employed residents held onshore oil jobs in 2003, in the municipality or by commuting (Section 3.2).

Shelf workers could live far from the industry because their work leaves them long stretches at home. Since the Working Environment Act of 1977 was applied to the fixed platforms, limiting working hours for round-the-clock work, crews have worked two weeks on the platform followed by at least three weeks ashore (Meland, 2019b). The pattern became two weeks on, three off, two on and four off after 1986, and two weeks on and four off in 2002 and 2003, first by local agreements and then under the offshore collective agreements (Meland, 2019a; Knutsen et al., 2025). Today's collective agreements also make the employer pay for the trip between the worker's home and the heliport (Offshore Norge, 2024, s. 15.1). A worker who spends most of the year at the mainland does not need to live near the heliport, so the worker, in nine cases out of ten a man, and his family can live where they choose. Many did so long before the boom. In the census of 1980, 773 of the 2,192 residents who were at work in the North Sea in the census week, 35 percent, lived outside Rogaland and Hordaland. In every year from 2000, the first year of the register series, to 2017, between 45 and 50 percent of offshore workers lived outside Rogaland and Hordaland (Table 2).

The fall in the oil price that hit Stavanger should therefore have reached these regions too, through the earnings of residents who worked at the shelf. Shelf workers lost their jobs wherever they lived, including in towns with no oil firms at all. This paper asks how a shock that reaches a place only through its residents' wages affects its labour and housing markets. The question is not Norway's alone. Wherever people work far from where they live, a shock to a distant employer reaches their home town through their pay, and rotation work makes that exposure measurable. To answer it, I use public municipal data from Statistics Norway and a difference-in-differences

design in which municipalities differ in how much they were exposed. Exposure is the share of a municipality's employed residents who worked on the shelf in 2003, the year before the oil price began to rise, and it is held at that value throughout. By 2003 the rotation, two weeks offshore followed by three or four weeks at home, had long shaped where offshore workers lived. The sample is the 375 municipalities where at most 0.5 percent of employed residents held onshore oil jobs at the oil companies and their service firms in 2003. I count the jobs in the municipality and the jobs their residents commuted to elsewhere, so the bust could hardly reach them through oil firms, local or nearby. These municipalities held 7,134 (48% of the total) offshore workers. The design assumes that, had the oil price not fallen, municipalities with more offshore workers in 2003 would have followed the same path as the others.

Table 2: Residents whose workplace is the continental shelf, 1980 and 2000–2017

Year	Norway	Rogaland	Hordaland	Elsewhere	Elsewhere (%)
1980 (census week)	2,192	929	490	773	35.3
2000	15,477	4,969	3,605	6,903	44.6
2003	14,912	4,294	3,658	6,960	46.7
2005	16,290	4,842	3,842	7,606	46.7
2010	17,454	5,059	4,154	8,241	47.2
2013	20,603	5,609	4,840	10,154	49.3
2015	17,842	4,767	4,261	8,814	49.4
2017	14,742	4,035	3,624	7,083	48.0

Note: 1980: residents aged 16 and over who were at work in the week of 25–31 October 1980 and whose meeting place was the North Sea, from the full count of the 1980 census (table 9 of Statistics Norway's municipal census booklets). Workers ashore on their rotation that week are not counted, so the 1980 row is comparable with the later rows in its shares, not its levels. A booklet lists the North Sea only when it is among the municipality's main destinations. It is listed by 23 of the 26 municipalities of Rogaland, 19 of the 34 of Hordaland and 69 of the 394 elsewhere, so the municipalities left out lie mostly outside the two counties; the row sums the 110 whose numbers can be read. 2000–2017: employed residents whose workplace is the continental shelf in the fourth quarter, by county of residence on the 2019 map (Statistics Norway table 03321). The register series starts in 2000, and later years are not shown because Statistics Norway stopped recoding offshore establishments to the shelf after 2018. The 1990 census reports the North Sea only from a sample in municipalities with 6,000 or more residents and suppresses its small cells, so it is not shown.

Per percentage point of employed residents working offshore in 2003, the average income from employment fell by 1.9 percent between 2013 and 2017 relative to other municipalities, employment of residents by 1.4 percent and the average sale price of dwellings by 7%. The dwelling stock and the population did not move during the oil price fall. Prices fell about four times as much as income. With the coastal industries and the distance to a heliport held fixed the estimates are about two-fifths smaller, and a straight continuation of the 2009 to 2012 path would then cover zero (Section 6.2). The effects sit on the region where offshore workers live rather than on the municipality and the oil firms, the coastal industries and the equipment suppliers left in the sample do not explain them. The loss of income was the men's. Women's income did not fall during the period. For each offshore job lost, the equipment suppliers of the industry, which the sample rule does not remove, lost 1.3 jobs, and the rest of the local economy lost about two, although that estimate is imprecise. Whether laid-off offshore workers found other jobs cannot be seen in municipal

data, but employment outside oil fell rather than rose, so other industries did not add jobs on net.

Measuring the shock where the workers lived separates a shock to a region's workers from a shock to its firms, and it shows that lost wages move house prices in places with almost no oil firms. The firms that were hit, the equipment suppliers, lost jobs but do not explain the fall in prices.

2 Setting

Much of the oil wealth belongs to the state and goes into a sovereign fund, which held 6,431 billion kroner at the end of 2014. A fiscal rule lets the government spend about the expected real return of the fund each year, so a fall in the oil price lowers what flows into the fund and not what the budget can spend. The government leaned against the downturn with the budgets of 2015 and 2016, Norges Bank cut the policy rate from 1.5 percent to 0.5 percent between December 2014 and March 2016, and the krone lost about a fifth of its value against the import-weighted index (International Monetary Fund, 2016). These responses reached every municipality, and the comparison across municipalities removes them. The weaker krone helped the coastal exporters, and I hold their industries fixed in the robustness checks.

2.1 Offshore work

Norway's oil and gas lie under the continental shelf, and the onshore oil economy grew up around Stavanger and Bergen (Figure 1). In 2003 the extraction of oil and gas and the services around it employed 28,775 people, 16,593 of them on land, and the Stavanger/Sandnes region held 59 percent of the onshore jobs. The other 12,182 jobs of division 11 were on the shelf itself, which Statistics Norway records as a place of work of its own, and with the platform work and catering registered there as well, 14,912 residents had the shelf as their workplace (Section 3.1). Crews flew from five heliports, so the workplace fixes the heliport a worker flies from but not where the worker lives. Where they live is not random. The 2003 share correlates -0.53 with the log distance to the nearest heliport and 0.44 with the share working in ship and platform building, and the robustness checks hold these coastal traits fixed.

2.2 Boom and bust

The oil price averaged between 25 and 29 dollars a barrel from 2000 to 2003 and stayed around 110 from 2011 to 2013 (Figure 2). Investment on the shelf rose from 77 to 219 billion kroner of 2013, and the number of residents working on the shelf from 14,912 in 2003 to 20,603 in 2013. I measure the exposure in 2003, before the boom moved it. The bust began in June 2014, and the number of residents working on the shelf fell from 20,603 in 2013 to 14,742 in 2017. The cuts fell on exploration, drilling, engineering and supply, the activities that employ rotation crews, so the loss of offshore pay arrived from 2014 in every municipality that housed offshore workers.

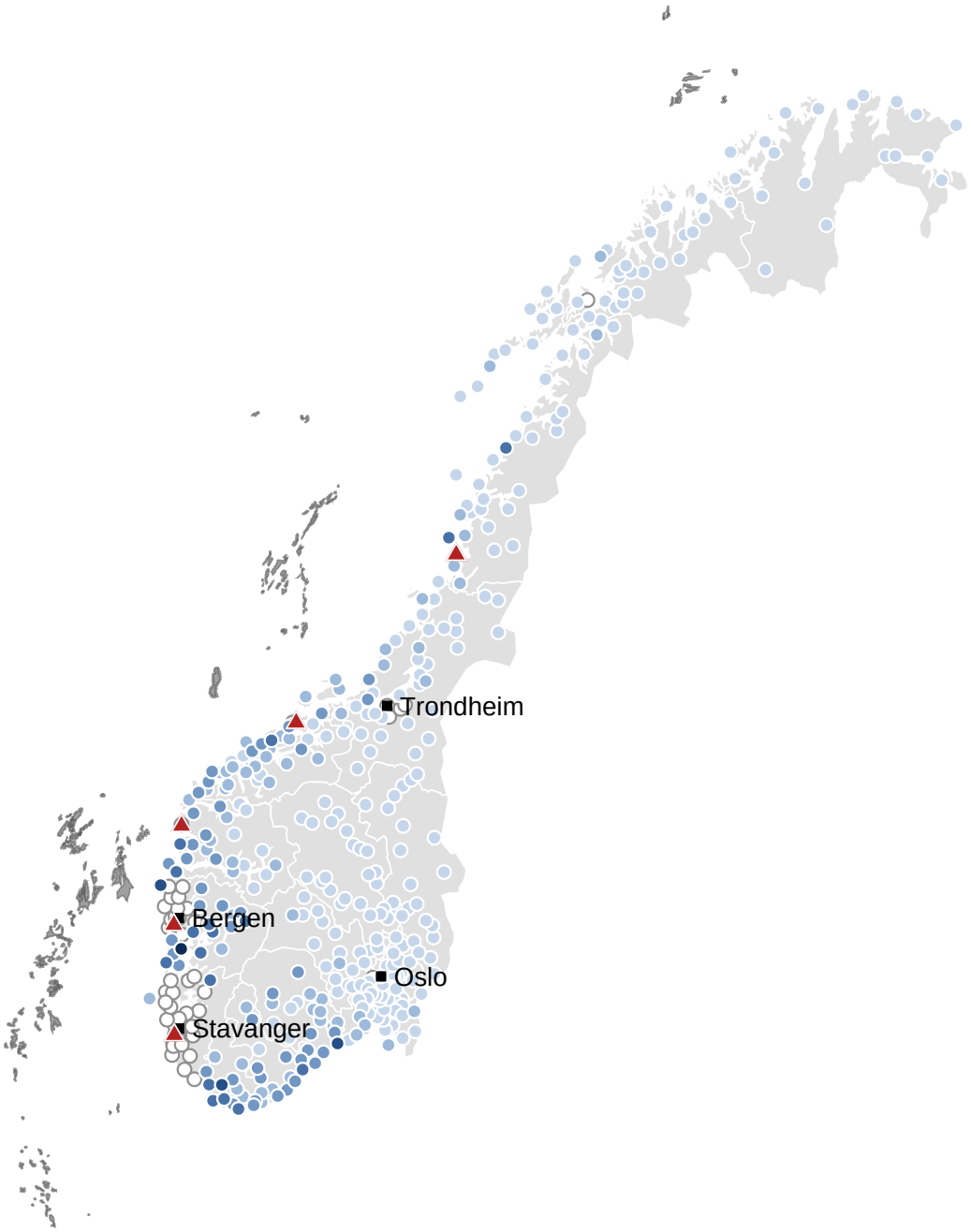
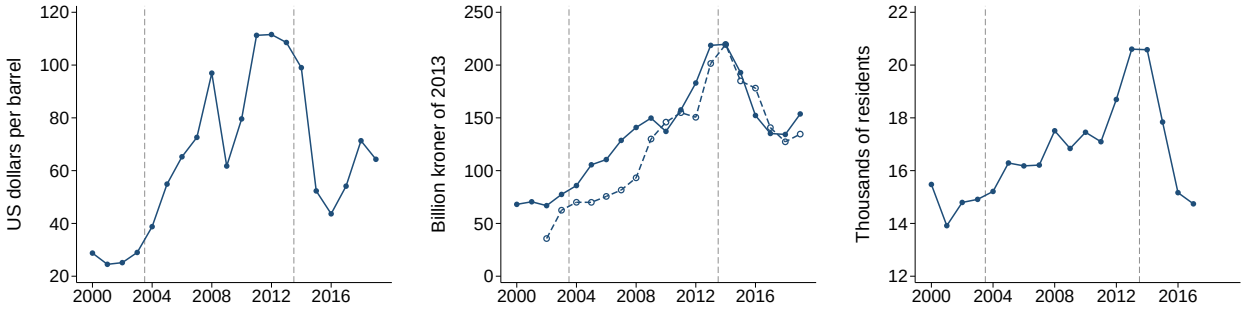


Figure 1: Oil and gas fields, offshore work and the sample

Note: The grey areas at sea are the outlines of the 139 oil and gas fields in the Offshore Directorate's FactMaps in October 2026, which include fields developed after 2003. Triangles mark the five heliports from which crews flew offshore in 2003: Stavanger (Sola), Bergen (Flesland), Florø, Kristiansund and Brønnøysund. Each point on land is a municipality, at its administrative centre. Points in the sample are shaded by the share of employed residents whose workplace is the continental shelf in the fourth quarter of 2003: from lightest to darkest below 0.5 percent, 0.5 to 1, 1 to 2, 2 to 3, 3 to 4, and 4 percent or more. White points are the 46 municipalities outside the sample, where more than 0.5 percent of employed residents held onshore oil jobs, in the municipality or by commuting (Section 3.2). Lines inside Norway are the 18 counties of 2019. Squares mark four towns. Tønsberg and Sandefjord, which are one unit in the data because Stokke was split between them in 2017, are both drawn.



(a) Oil price

(b) Planned and invested petroleum investment

(c) Residents working on the shelf

Figure 2: The boom and the bust, 2000–2019

Note: Panel (a) is the annual mean of the daily Brent spot price in US dollars per barrel (US Energy Information Administration, through FRED). Panel (b) shows accrued investment in oil and gas extraction and pipeline transport (solid line, filled markers; Statistics Norway) and the investment that the oil companies had planned for the same year in the May survey of the year before (dashed line, hollow markers), in billion kroner of 2013, deflated with the consumer price index. Panel (c) is the number of residents whose workplace is the continental shelf in the fourth quarter, shown to 2017 (Section 3.1). The dashed lines fall between 2003 and 2004 and between 2013 and 2014.

2.3 Layoffs and benefits

A worker who lost an offshore job received the same unemployment insurance as any other worker, 62.4 percent of previous earnings up to a cap of six times the base amount, for up to two years. Employers could also lay workers off temporarily, and the employment register counts a worker on temporary layoff as employed if the layoff is expected to last less than 90 days. The income measure leaves benefits out, so it records the loss of pay before insurance.

3 Data

All data used in this paper are retrieved from public sources of Statistics Norway's StatBank, the Labour and Welfare Administration (NAV) and the Norwegian Offshore Directorate.[1] Municipality level data is used in the regressions and each year's data is carried forward to 2019 with Statistics Norway's official list of code changes. If municipalities merged during the sample period, the merged unit carries the whole population throughout the sample. The only split between 2000 and 2019 is that of Stokke in 2017, so I merge its two successors (Tønsberg and Sandefjord) into one unit. This leaves 421 municipalities in 88 economic regions.

3.1 Exposure

Statistics Norway records the continental shelf as a place of work of its own and links each worker to the municipality where they live. For each municipality I divide the number of residents whose workplace is the shelf by the number of employed residents. In 2003, the workplace of 14,912

[1] Table 11 in the appendix lists every table.

Norwegian residents was registered as the shelf. 82 percent of them worked in the extraction of oil and gas and the services around it[2], 10 percent in platform and yard work and 6 percent in catering. I fix the exposure in 2003, the last year before the oil price began to rise, so exposure is the share of a municipality's employed residents whose workplace was the continental shelf in the fourth quarter of 2003.

The count of shelf workers is a lower bound. Employers register their establishments at fixed addresses, and Statistics Norway moved a job to the shelf only when it knew the establishment worked offshore. These jobs are mainly in oil services, drilling and catering. Offshore workers of employers registered at an onshore office stay at that office's address, workers on mobile rigs are recorded as seafarers. Foreign residents and commuters from Sweden have no home municipality at all. If every municipality's count misses the same fraction of its offshore workers, they overstate the effect of one true point by that fraction. If the fraction varies across municipalities for reasons unrelated to the outcomes, it weakens them. Exposures calculated from 2000 data correlates 0.93 with its value in 2003. Therefore, the shares are assumed to be stable before the boom. The series changes source in 2015 and breaks again in 2018, both well after the year I use to calculate the exposure.

In the 2003, the share of employed residents working offshore was zero in 40 municipalities and 0.28 percent at the median (Table 4). It was 1.33 percent at the 90th percentile and above 3 percent in four municipalities (Table 1). Weighted by population, the mean is 0.41 percent and the standard deviation 0.57. The coefficients in this paper are per percentage point of exposure, which is about the difference between the 90th and the 10th percentile. Figure 1 maps the share across Norway.

3.2 Sample

The sample is restricted to municipalities whose residents held almost no onshore oil jobs in 2003 since the design needs places that the bust could reach only through their residents. For each municipality I count its employed residents by the municipality where they work and weight each of them by the share of oil extraction and related jobs among all jobs located in that workplace municipality. The sum includes the residents who work in their own municipality, so the measure covers oil firms in the municipality as well as commuting to oil jobs elsewhere. Divided by the municipality's employed residents, it estimates the share of its workers who held onshore oil jobs, on the assumption that a commuter holds an oil job as often as the average worker where they work. Jobs on the continental shelf are left out, since they are the exposure. Across all municipalities the measure allocates the 16,593 onshore oil jobs to the places where their holders lived. The cutoff is set by a gap in the data. In 344 of the 421 municipalities the share was at most 0.2 percent, and in the median one it was 0.07 percent. Above 0.2 percent the values rise without a break to 0.48 percent in Porsgrunn, and the next municipality, Etne, has 0.57 percent. I keep the municipalities at or below 0.5 percent. The larger gaps further up, above 1.2 percent, would keep the oil-office cities and their commuter suburbs, such as Bærum, Trondheim, Askøy and Karmøy. The rule leaves out

[2](division 11 of the industry classification SN2002)

the oil centres of Stavanger, Haugesund and Bergen with their commuter belts which corresponds to 46 municipalities. The sample is the remaining 375 municipalities in 86 economic regions. They had 3.8 million inhabitants in 2014 (75% of the population), and 7,134 of the 14,912 offshore workers (48%).

A rule based on the oil jobs located in the municipality would not do the same job. It would keep 30 municipalities that this rule leaves out, most of them suburbs and islands around Stavanger, Haugesund and Bergen, such as Gjesdal, Rennesøy, Strand, Klepp and Time. They have few oil offices of their own, but their residents commute to the offices next door. In these 30 municipalities, 1.5 percent of employed residents held onshore oil jobs, against 0.1 percent in the sample, and 2.3 percent worked offshore, more than five times the sample's 0.41. Including them would put the places most exposed to the onshore oil economy at the top of the offshore distribution.

Division 11 is a narrow definition of the oil jobs. It leaves out the yards, the equipment makers, the supply ships and the supply bases, which were also hit in 2015. I therefore build a second rule for the supply chain, measured in the same year as the exposure. Statistics Norway's list of petroleum-related industries exists only from 2008, five years into the boom, but almost all of its jobs, 90 percent in 2008, fall into two divisions of the older classification: division 11 and division 35, which covers ships and platforms.[3] Both divisions are published by place of work for 2003 and their share of a region's jobs in 2003 correlates 0.98 with the share on the 2008 list. Table 3 shows this rule and the other samples I use to test the choice of sample, and Section 6 reports the estimates on each.

Table 3: The headline sample and the alternative samples

			Population	Offshore workers		Offshore work (%)	
Sample	Munic.	Regions	(1,000)	Number	% of all	Mean	P90
All municipalities	421	88	5,109	14,912	100	0.67	2.04
Headline sample	375	86	3,849	7,134	48	0.41	1.12
Onshore jobs located in the region only	360	79	3,778	7,358	49	0.44	1.18
Outside Rogaland and Hordaland	362	80	4,144	6,960	47	0.38	0.89
No onshore oil job in the region	259	59	1,721	3,138	21	0.40	0.98
Regional onshore share $\leq 1.2\%$	381	84	4,285	8,211	55	0.43	1.18
Municipal onshore share $\leq 0.5\%$	404	88	4,089	9,513	64	0.53	1.44
Supply chain 2003 $\leq 1\%$	256	56	3,257	3,942	26	0.27	0.77
Supply chain 2003 $\leq 2\%$	296	66	3,506	4,998	34	0.32	0.78

Note: Population on 1 January 2014. Offshore workers are residents whose workplace was the continental shelf in the fourth quarter of 2003, and offshore work is their share of employed residents; the mean and the 90th percentile are weighted by population. The rules are those of Section 3.2: the headline rule counts the onshore oil jobs held by a municipality's residents, the regional rule the onshore oil jobs located in its region, and the supply-chain rule the region's jobs in SN2002 divisions 11 and 35, all in 2003.

[3](Statistics Norway's correspondence table between SN2007 and SN2002)

3.3 Outcomes

The outcomes follow the shock from income to the labour market and to housing (Table 4). The first stage is the average income from employment per resident aged 17 and over, from the tax records from 2005 onwards. It leaves benefits out, so it measures the loss of pay before insurance and taxes, and because it is measured per resident it moves with both the pay of those in work and the share of adults who work. When municipalities merged, I average their means with population weights. To see whose income fell, I also use it for men and for women separately, since nine in ten of the shelf jobs of 2003 were held by men. As a check I use the median household income after tax, which includes unemployment benefits and so measures the loss after insurance and taxes (Section 2.3), averaged with household weights when municipalities merged.

For the labour market I use the number of employed residents aged 20 to 66 in the fourth quarter and the registered unemployment rate in November. The employment register moved to a new source in 2015. Statistics Norway gives no overlap year or back-cast by residence, but it published the 2015 count of employees by workplace municipality twice, on the new source and adjusted to the old one. The gap between the two, the size of the break in each municipality, does not follow offshore work (Section 6.3), so I measure employment from 2013 to 2017, like income and prices.

To see which jobs were lost, I also use employed residents by industry of their workplace from 2013 to 2019.

For housing I use the average price per sale of dwellings sold on the free market, the dwelling stock on 1 January and the number of dwellings given a building permit. The average price is total purchase value divided by the number of sales, so I use it only where a municipality had at least 10 sales in every year from 2003 to 2019, which keeps the panel balanced. That leaves 271 of the 375 municipalities, with 96 percent of the sample's population and 95 percent of its offshore workers. The average price is not adjusted for the size or type of the dwellings sold, so I also report the price per square metre of detached houses, which Statistics Norway publishes only where there were enough sales and which is observed in every year in 135 municipalities. The population on 1 January comes from table 06913.

The bust windows start in 2013, the last year before the price fell. Income, employment and prices are measured from 2013 to 2017, registered unemployment from November 2013 to November 2016, and the dwelling stock and the population from 1 January 2014 to 1 January 2018. For the boom the windows run from 2003 to 2013, or from 2005 for income and from 2006 for the dwelling stock, where those series begin. Every outcome is a long difference, 100 times the change in logs, except for registered unemployment, which is the change in percentage points.

3.4 Controls

Every regression holds fixed the size of the municipality and its industry mix before the boom. These are the log population at the start of the window and the 2003 shares of employed residents in industry and mining, in the primary industries and in trade, hotels and restaurants. It also holds fixed two measures of the onshore oil economy: the onshore oil share of the jobs in the municipality's

Table 4: Summary statistics, headline sample

			Headline sample				All
Variable	Unit	*N*	Mean	P10	Median	P90	Mean
Exposure and sample rules, 2003							
Offshore work	% employed residents	375	0.41	0.00	0.28	1.33	0.67
Onshore oil economy of the region	% of the region's jobs	375	0.07	0.00	0.00	0.03	0.75
Residents' onshore oil jobs	% employed residents	375	0.10	0.02	0.06	0.19	0.77
Supply chain of the region	% of the region's jobs	375	0.98	0.01	0.35	3.62	2.09
Controls							
Fishing and fish farming	% employed residents	375	0.83	0.00	0.19	8.86	0.72
Ship and platform building	% employed residents	375	1.02	0.05	0.42	3.29	1.37
Water transport	% employed residents	375	0.88	0.08	0.56	3.41	0.97
Distance to the nearest heliport	km	375	261.82	58.29	192.07	445.19	215.76
Population, 1 January 2014	thousands	375	10.26	1.15	4.33	20.60	12.14
Outcomes, bust							
Mean employment income per resident, 2013–17	log points	375	8.76	4.24	9.11	12.89	7.76
Median household income after tax, 2013–17	log points	375	9.71	5.80	9.67	13.43	9.00
Employed residents, 2015–17	log points	375	2.07	−3.24	0.60	3.91	1.80
Registered unemployment, Nov 2013–16	pp	373	−0.11	−1.20	−0.10	1.08	0.26
Average price per sale, 2013–17	log points	271	27.68	8.24	26.27	46.41	24.15
Detached-house price per m^2, 2013–17	log points	225	22.48	5.36	20.45	31.80	18.92
Dwelling stock growth, 2014–18	log points	375	4.66	0.59	3.57	7.38	4.96
Population growth, 2014–18	log points	375	3.42	−3.53	0.74	5.40	3.55

Note: Municipalities in the headline sample (375) and in Norway (421). Means are weighted by population on 1 January 2014, except for population itself; the percentiles are across municipalities. Log points are 100 times the change in logs and pp are percentage points. The average price per sale is shown for the 271 municipalities of the balanced panel, and the detached-house price where Statistics Norway publishes one. The supply chain of the region is jobs located in the region in SN2002 divisions 11 and 35, as a share of all jobs there, in 2003.

region, which matters for the sample municipalities that lie in oil regions, and the share of its own residents who held onshore oil jobs, the measure behind the sample rule. Offshore workers live near the heliports and in communities that already worked at sea (Section 2.1), so a second set of controls measures those traits in 2003. These are the shares of employed residents in fishing and fish farming, in ship and platform building and in water transport, and the log distance from the municipality's administrative centre to the nearest of the five heliports in use in 2003.

A third set, used in the robustness checks, holds fixed the distance to the nearest of Oslo, Bergen, Trondheim and Stavanger and the 2008 share of residents working in fabricated metal products and in machinery and equipment, the industries that supplied the oil industry.

4 Empirical strategy

Δy_i is the change in the outcome over the bust window in municipality i and S_i is the share of its employed residents who worked offshore in 2003. W_i holds the controls of Section 3.4, the two onshore oil measures, the log population and the 2003 industry shares. The regressions are weighted by population at the start of the window, so β is the effect for the average resident per percentage point of offshore work.

$$\Delta y_i = \alpha + \beta\, S_i + W_i'\gamma + \varepsilon_i \tag{1}$$

4.1 Event studies

The outcome y_{it} is the change from 2013 to year t, so β_k after 2013 is the cumulative effect of the bust up to year k, and every control is interacted with year. The coefficients before 2013 test whether exposed municipalities were already diverging from the others before the bust.

$$y_{it} = \lambda_t + \sum_{k \neq 2013} \mathbf{1}[t = k]\,(\beta_k\, S_i + W_i'\gamma_k) + \varepsilon_{it}, \qquad y_{i,2013} = 0 \tag{2}$$

4.2 Region and municipality

$\bar{S}_r$ is the population-weighted mean of S over the sample municipalities of region r, the economic region of municipality i. β_R is the effect of living in a region with more offshore workers and β_W the extra effect of having more of them than the neighbouring municipalities.

$$\Delta y_i = \alpha + \beta_R\, \bar{S}_r + \beta_W\,(S_i - \bar{S}_r) + W_i'\gamma + \varepsilon_i \tag{3}$$

4.3 The 2013 share

The 2013 share S_i^{13} measures the dose at the time of the bust, but where offshore work grew during the boom depends on how places fared in it. I therefore instrument it with the 2003 share, so β^{IV}

is the effect per point of offshore work when the bust began, identified from the part of the 2013 share that was already in place in 2003.

$$
\begin{aligned}
S_i^{13} &= \pi\, S_i + W_i'\theta + u_i, \\
\Delta y_i &= \alpha + \beta^{IV}\, \widehat{S}_i^{13} + W_i'\gamma + \varepsilon_i
\end{aligned}
\tag{4}
$$

4.4 Inference

Standard errors are clustered by the 86 economic regions of the sample. Every p-value and star comes from a wild cluster bootstrap-t over regions with Webb weights and 9,999 draws (Roodman et al., 2019), and the main table also reports Romano–Wolf p-values for the six bust outcomes tested together.

4.5 Identification

The identifying assumption is that, given the controls, municipalities with more offshore workers in 2003 would have followed the same path after 2013 as the others had the oil price not fallen. With one national shock, identification comes from the shares (Goldsmith-Pinkham et al., 2020). Four threats matter. The first is the oil industry left in the sample, which the sample rule, the onshore controls and Section 5.2 address. The second is coastal geography and the krone, which the coastal controls address. The third is that peripheral places follow the national housing cycle less closely, and the fourth is that the only years before the bust are boom years, so a trend before 2013 would be hard to tell from the bust. The event studies show no such trend, and Section 6 takes up the last two. The 2003 share predicts none of the outcomes before the boom, neither population growth in the 1980s and 1990s nor the change in employment and registered unemployment from 2000 to 2003, on the headline sample and on the samples of the robustness checks.[4]

5 Results

Table 5 reports the bust. Per percentage point of offshore work, the average income from employment fell by 1.9 log points between 2013 and 2017, employment of residents by 1.4 and the average sale price of dwellings by 7.1. The dwelling stock and the population did not move. The income, employment and price effects survive the Romano–Wolf correction for six outcomes. A municipality at the 90th percentile of exposure has 1.3 points more offshore workers than one at the 10th, so the estimates imply an income about 2.6 percent lower and prices about 9.5 log points lower in the more exposed of the two. Panel B gives the effect per point of offshore work in 2013, and it is smaller by about a tenth.

Figure 3 shows the timing. Income, employment and prices are flat relative to 2013 from 2009 to 2012. Prices start to fall in 2014 and keep falling to 2018, employment falls in 2015 and 2016,

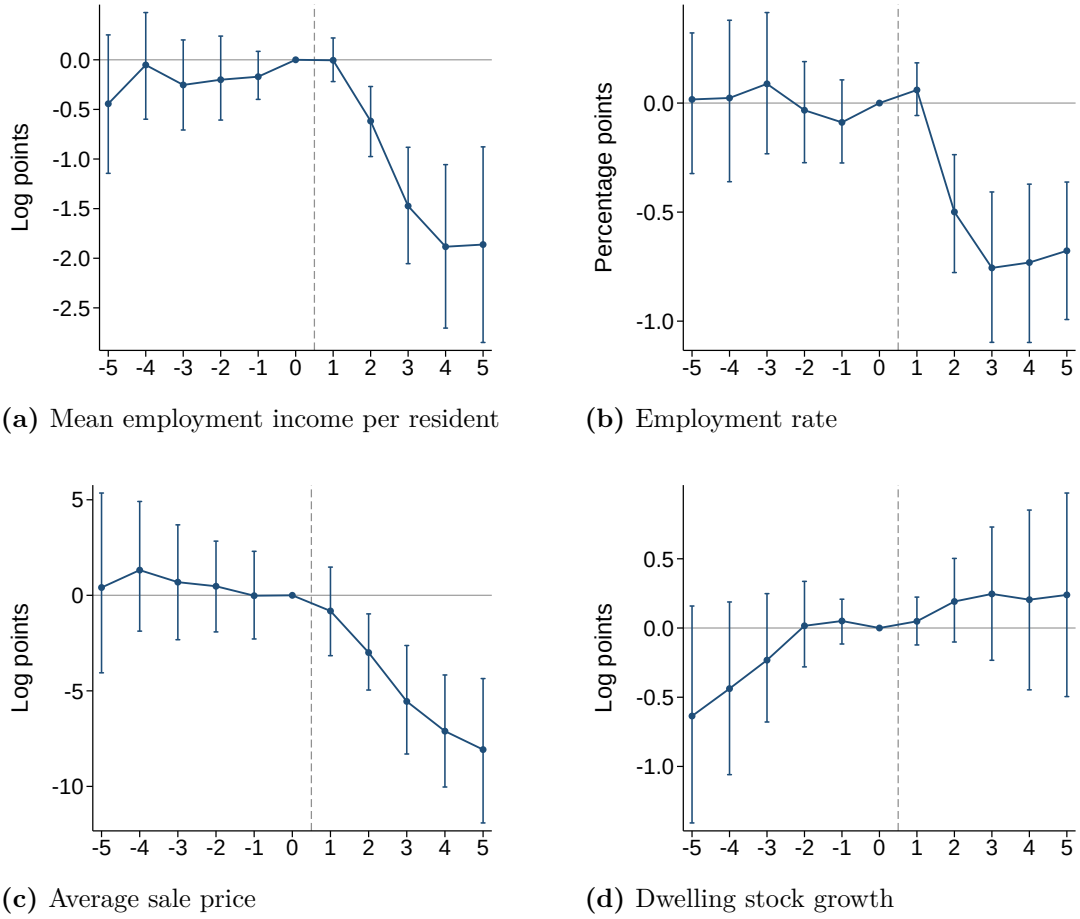
[4]Table 17. One of the 24 coefficients, employment growth from 2000 to 2003 on the narrower supply-chain sample, has a p-value of 0.05.

Table 5: The bust where offshore workers live, 2013–2017

Dependent variable	Income (1)	Employment (2)	Price (3)	Dwelling stock growth (4)	Population (5)
Panel A. Offshore work in 2003					
Offshore work, 2003	−1.932***	−1.378***	−7.142***	0.226	−0.284
	(0.385)	(0.386)	(1.444)	(0.302)	(0.272)
Wild bootstrap p-value	<0.001	0.002	<0.001	0.506	0.318
Romano–Wolf p-value	0.002	0.017	0.002	0.553	0.553
Municipalities	375	375	271	375	375
Panel B. Offshore work in 2013, instrumented by offshore work in 2003					
Offshore work, 2013, instrumented	−1.712***	−1.221***	−6.400***	0.201	−0.252
	(0.374)	(0.388)	(1.435)	(0.267)	(0.248)
First stage, 2013 on 2003 share	1.13	1.13	1.12	1.13	1.13
First-stage F (Kleibergen–Paap)	342	342	337	342	342
Municipalities	375	375	271	375	375
Onshore oil controls	✓	✓	✓	✓	✓
Population and industry controls	✓	✓	✓	✓	✓
Population weights	✓	✓	✓	✓	✓

Note: Coefficients on offshore work in 2003, in percentage points of employed residents, in the 375 municipalities of the headline sample (Section 3.2). The outcomes are the bust long differences of Section 3.3; the price is on the 271 municipalities of the balanced panel, and employment runs across the 2015 change of register source (Section 6.3). Controls, weights and inference as in Sections 3.4 and 4.4: standard errors clustered by economic region in parentheses, stars and p-values from the region wild bootstrap, * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, and Romano–Wolf p-values over six outcomes, these five and the change in registered unemployment. Panel B estimates Equation (4) by two-stage least squares; its stars come from the wild restricted efficient bootstrap over regions and the first-stage F is the Kleibergen–Paap statistic with region clusters.

and income follows from 2015. All three are still down in 2018.



(a) Mean employment income per resident

(b) Employment rate

(c) Average sale price

(d) Dwelling stock growth

Figure 3: Offshore work, income, employment and housing, five years either side of 2013

Note: Coefficients β_k of Equation (2) per percentage point of offshore work in 2003, with 95 percent region wild bootstrap intervals, in the 375 municipalities of Table 5. The horizontal axis counts years from 2013, so -5 is 2008 and 5 is 2018, and the dashed line falls between 2013 and 2014. Income is 100 times the log difference from 2013 of the average income from employment per resident. The employment rate is the change from 2013 in percentage points and changes source in 2015. The average sale price is on the 271 municipalities of the balanced panel, and dwelling stock growth is 100 times the log difference of the end-of-year stock from 2013.

5.1 Regional effect

Four-fifths of the variation in exposure lies between economic regions, and so do the effects (Table 6). The price effect sits on the region's mean exposure, at -9.9 log points per point, and a municipality with more offshore workers than its neighbours shows no larger fall. Region fixed effects remove it. The income effect is mostly regional too, with a within-region part about a third as large.

Table 6: Region mean and own exposure, bust 2013–2017

Dependent variable	Income (1)	Employment (2)	Price (3)
Panel A. Region mean and own deviation			
Offshore work, region mean	−2.342***	−1.633***	−9.870***
	(0.443)	(0.420)	(1.622)
Offshore work, own deviation	−0.625*	−0.565	2.034
	(0.324)	(0.468)	(2.339)
Municipalities	375	375	271
Panel B. Economic-region fixed effects			
Offshore work, 2003	−0.717*	−0.705	1.643
	(0.410)	(0.523)	(2.999)
Municipalities	375	375	271

Note: Panel A splits offshore work into the region mean, the population-weighted mean over the region's sample municipalities (weights from 1 January 2004), and the municipality's deviation from it (Equation (3)). Panel B replaces the region mean by economic-region fixed effects, which absorb the region's onshore oil job share. Outcomes, controls, weights, inference and stars as in Table 5.

5.2 Supply chain and coast

The oil industry left in the sample does not explain the regional effect (Table 7). The price effect stays between −6.7 and −10.6 log points when the coastal industries, the heliport distance, the supply-chain jobs, the equipment suppliers and the distance to Stavanger and Bergen are held fixed, and when the regions with a supply-chain hub are left out. The income effect loses about a third with the coastal controls and holds in the other columns except with the distance quintiles. The employment effect does not survive the stricter exclusion of hub regions.

5.3 Who lost

The income loss fell on men, who held nine in ten of the shelf jobs. Income per adult man fell by 3.1 log points per point of exposure and per adult woman by 0.0, which cannot be told from zero. For men the loss splits between fewer men in work, 1.2 log points by 2017, and lower pay per man in work, 1.8. The offshore jobs actually lost were 0.35 per 100 employed residents per point, while total employment fell by 1.4, and no industry outside oil gained except agriculture, forestry and fishing. Not all of the rest is a multiplier. The equipment suppliers of the industry, machinery and metal fabrication and the support codes on Statistics Norway's petroleum list, lost 0.45 jobs per 100, 1.3 for each offshore job. That is a direct hit on local firms, because the sample rule removes the oil companies and their service firms but not their suppliers. The rest of the local economy, mainly hotels, restaurants and transport, lost 0.64 jobs per 100 ($p = 0.10$), 1.9 for each offshore job, with a 95 percent interval from −0.1 to 3.8. Removing all of manufacturing instead leaves 1.5. The multiplier on offshore pay is therefore about two, not the three to four that the raw count outside oil suggests, and it is not precisely estimated. Dropping the regions with yards and other supply-chain jobs does not remove the suppliers' loss. On the two supply-chain samples of Table 3

Table 7: The regional effect against oil-industry and coastal controls, bust 2013–2017

	Baseline (1)	Coastal (2)	Horse race (3)	Hubs 5% (4)	Hubs 3% (5)	Distance (6)	Suppliers (7)
Panel A. Mean employment income							
Offshore work, region mean	−2.342***	−1.482***	−1.949***	−1.997***	−1.660**	−1.260	−1.538***
	(0.443)	(0.443)	(0.559)	(0.548)	(0.534)	(0.600)	(0.406)
Municipalities	375	375	375	272	240	375	375
Panel B. Employment, 2013–17							
Offshore work, region mean	−1.633***	−1.612***	−1.762***	−1.388*	−0.704	−1.317**	−1.573***
	(0.420)	(0.455)	(0.554)	(0.755)	(0.691)	(0.423)	(0.453)
Municipalities	375	375	375	272	240	375	375
Panel C. Price							
Offshore work, region mean	−9.870***	−7.882***	−10.552***	−8.952**	−7.997*	−6.704**	−7.890***
	(1.622)	(1.935)	(2.043)	(2.531)	(2.958)	(2.378)	(1.922)
Municipalities	271	271	271	197	176	271	271
Coastal controls		✓	✓	✓	✓	✓	✓
Other oil measures			✓				
Hub regions left out				✓	✓		
Distance quintiles						✓	
Supplier exposure							✓

Note: Coefficients on the region mean of offshore work in 2003, in percentage points of employed residents. Every regression includes the municipality's deviation from the region mean and the controls, weights and inference of Table 5. Column (1) is panel A of Table 6. Column (2) adds the coastal controls of Section 3.4 and the region's share of petroleum-related jobs in 2008. Column (3) also adds the share of petroleum-related jobs located in the municipality in 2008 and the share of onshore oil workers among its residents in 2003, each as the municipal value and the region mean. Columns (4) and (5) use the controls of column (2) and leave out the regions that contain a hub, a municipality with at least 5 or 3 percent of its jobs on Statistics Norway's petroleum-related list. Column (6) adds four dummies for quintiles of the distance to the nearer of Stavanger and Bergen. Column (7) adds the 2008 supplier share of Section 3.4, as the municipal value and the region mean. Stars as in Table 5.

it is 0.50 and 0.64 jobs per 100, larger than on the headline sample, because the firms that lost the jobs make machinery and metal parts and sit outside the yard regions, while the loss in the rest of the local economy is 0.9 and 0.5 jobs per 100, three and one and a half times the offshore jobs, with intervals that include zero.[5] Valued at the pay of the oil industry, the offshore jobs lost explain about a quarter of the fall in employment income.[6]

Households were compensated for part of the loss. Per point of exposure, the average income from employment per adult fell by 1.9 log points, while gross income per adult, which adds benefits, pensions and capital income for the same people, fell by 1.4, so transfers and other income replaced about a quarter of the lost pay. The median household income after tax fell by 1.1. An offshore worker who lost the job received 62.4 percent of previous earnings for up to two years (Section 2.3), and the benefit records show it. Per point of exposure, the number of residents who received unemployment benefits during the year was 0.5 per 100 residents higher in 2016 and 2017 than in other municipalities and 0.2 higher in 2018, about two recipients for every three jobs lost, and the benefits paid were 360 kroner per resident higher in 2016, less than a tenth of the lost pay. The oil jobs of residents also partly came back in 2018 and 2019. Waiting at home was therefore a reasonable choice, which fits the unchanged population and the small net movement into other jobs. Prices fell all the same, so the housing market priced the loss of pay and not the insured income.

6 Robustness

Table 8 adds the distance to the nearest large city, which costs about an eighth of the price effect and a twentieth of the income effect, and then the coastal controls and distance together, which leave a price effect of -4.1 and an income effect of -1.1. The median household income after tax, which includes benefits, gives -1.1 against -1.9 for the mean, and the price per square metre of detached houses gives larger effects on its smaller balanced panel. One observation per region gives the same effects as the region mean in Table 6. Figure 4 shows that the paths before 2013 stay flat with these controls.

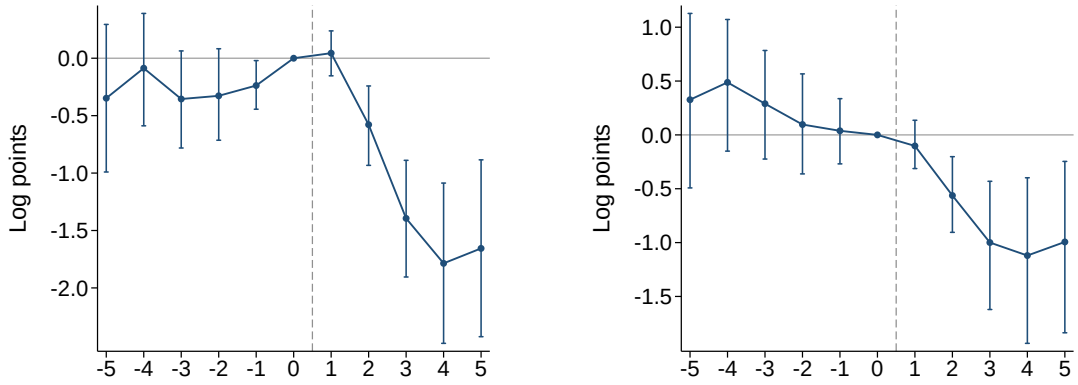
[5] Tables 12 and 13 report the change in employed residents by industry of workplace, by industry group and for the petroleum-related industries and the manufacturing divisions. Table 14 reports the decomposition and the ratios, and Table 15 repeats them on the supply-chain samples.

[6] The offshore jobs lost are 0.35 per 100 employed residents per point of exposure. Earnings per job in extraction and its services were 1.57 times the all-industry average in the fourth quarter of 2016 (Statistics Norway table 12314), so the direct loss is $0.35 \times 1.57 = 0.55$ percent of aggregate employment earnings, against the fall of 1.9 percent in employment income per adult, a share of 0.28. It is a lower bound, because it counts every lost job as a full year of pay lost and ignores pay cuts for those who kept their jobs.

Table 8: The bust with rural controls and other measures of income and price, 2013–2017

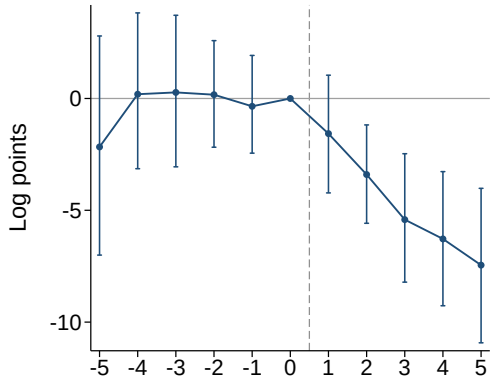
Dependent variable	Income (1)	Employment (2)	Price (3)
Panel A. Added controls			
Baseline	−1.932***	−1.378***	−7.142***
	(0.385)	(0.386)	(1.444)
Distance to the nearest large city	−1.824***	−1.106***	−6.219***
	(0.328)	(0.345)	(1.450)
Coastal controls and distance	−1.139***	−1.056***	−4.127**
	(0.346)	(0.359)	(1.699)
Panel B. Other measures of income and price			
Median household income after tax	−1.088***	–	–
	(0.218)		
Municipalities	375		
Detached price per m^2, balanced 2003–19	–	–	−10.285***
			(1.921)
Municipalities			135
Detached price per m^2, every municipality with a price	–	–	−8.302***
			(1.438)
Municipalities			225
Average price per sale, every municipality with 10 sales	–	–	−6.852***
			(1.428)
Municipalities			313
Panel C. One observation per economic region			
Offshore work, region mean	−2.310***	−1.353**	−10.241***
	(0.513)	(0.468)	(1.944)
Regions	86	86	86

Note: Coefficients on offshore work in 2003, in percentage points of employed residents, with the specification, weights and inference of Table 5. Panel A adds to the baseline the distance to the nearest large city and then the coastal controls of Section 3.4. Panel B uses the other income and price measures of Section 3.3: the median household income after tax, the detached-house price per square metre on the municipalities with a price in every year from 2003 to 2019 and on every municipality with a price in 2013 and 2017, and the average price per sale on every municipality with at least 10 sales in 2013 and 2017. In panel C each economic region is one observation, the population-weighted mean of the outcome over its sample municipalities, regressed on the region mean of offshore work, the region's onshore oil job share, log population and the region means of the 2003 industry shares, weighted by region population, with heteroskedasticity-robust standard errors and the wild bootstrap.

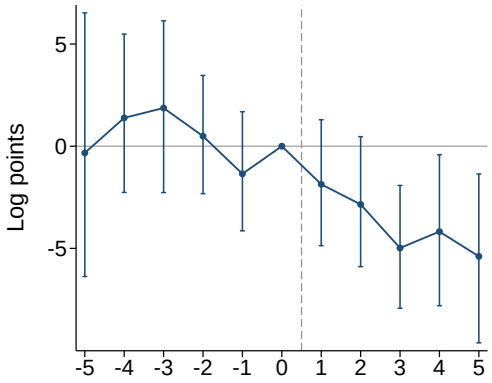


(a) Income, distance control

(b) Income, coastal controls and distance



(c) Price, distance control



(d) Price, coastal controls and distance

Figure 4: Income and price with the distance and coastal controls, five years either side of 2013

Note: The event studies of Figure 3 for the mean employment income per resident and the average sale price, with the controls of the second and third rows of panel A of Table 8 added, each interacted with year. The horizontal axis counts years from 2013. Points, intervals, weights and samples as in Figure 3.

6.1 The cycle

Table 9 fits one coefficient on national offshore employment over the boom and the bust. Income moved with it in the early boom and in the bust but not from 2009 to 2012, so the fall after 2014 is larger than a reversal of the late boom. Prices did not follow offshore employment in the boom. Exposed places follow the national house-price cycle less closely, and holding that fixed makes the bust price effect larger.

$$y_{it} = \lambda_t + \beta\, S_i Z_t + \kappa\, S_i h_t + \sum_{k \neq 2013} \mathbf{1}[t = k]\, W_i' \gamma_k + \varepsilon_{it}, \qquad Z_t = \frac{N_{2013} - N_t}{N_{2013} - N_{2017}} \tag{5}$$

Table 9: One coefficient on national offshore employment, 2003–2017

	Mean employment income		Average sale price	
	(1)	(2)	(3)	(4)
Whole cycle	−1.45***	−1.43***	−3.37**	−4.37***
2003–2008	−1.88***	−0.40	−3.04	2.75
2009–2012	−0.30	0.53	1.28	4.11**
2014 onward	−1.70***	−2.22***	−6.57***	−8.36***
Offshore work × national house prices		0.001		−0.032*
Proportional path, p-value	0.002	0.002	0.059	0.056
National house-price term		✓		✓

Note: Coefficients β of Equation (5), per percentage point of offshore work in 2003 and per unit of the 2013–2017 fall in national offshore employment, from the event-study sample of Figure 3 for 2003 (income 2005) to 2017. The rows for periods estimate β separately for 2003–2008, 2009–2012 and 2014–2017 in one regression. Columns (2) and (4) add offshore work times the national house-price index (κ). The last row is the p-value of the joint test that the event-study coefficients equal βZ_t. Controls, weights and inference as in Figure 3.

6.2 Pre-trends

Table 10 asks how far the path of 2009 to 2012 could continue after 2013 before the 2017 effect loses significance (Rambachan and Roth, 2023). With the baseline controls both the price and the income effect survive a straight continuation of the trend. With the coastal controls they do not.

6.3 Further checks

Five further checks are in the appendix.[7] The average sale price shows no trend from 1992 to 2002 with the baseline controls and a borderline one with the coastal controls, with joint p-values of 0.30 and 0.05. The rise of prices in exposed places in 2008 and 2009 goes with a crisis cushion in men's earnings, which held up in those two years relative to the surrounding ones, and it shows in the detached-house index rather than in the average sale price. The 2013 share by least squares

[7] Tables 18 to 20 and Figures 5 and 6.

Table 10: Sensitivity of the 2017 effect to the trend before 2013

	Average sale price		Mean employment income	
	Baseline (1)	Coastal (2)	Baseline (3)	Coastal (4)
Coefficient, 2017	−7.11	−5.16	−1.88	−1.18
95% interval	[−9.9, −4.3]	[−8.5, −1.8]	[−2.6, −1.2]	[−1.9, −0.5]
95% interval, 2009–13 trend continues ($M = 0$)	[−9.6, −1.8]	[−7.1, 3.5]	[−2.7, −1.1]	[−1.5, 0.3]
Breakdown M, smoothness	0.25	0.00	0.25	0.00
Breakdown $\bar{M}$, relative magnitudes	0.75	0.25	1.25	0.75

Note: Bounds of Rambachan and Roth (2023) for the 2017 coefficient of Equation (2), using the 2009–2012 coefficients as the pre-period. The smoothness restriction lets the 2009–2013 trend continue after 2013, with its slope changing by at most M a year; $M = 0$ is a straight continuation. The relative-magnitudes restriction bounds each yearly violation after 2013 by $\bar{M}$ times the largest yearly movement in 2009–2013. The breakdown value is the smallest M or $\bar{M}$ on the grid at which the 95 percent interval contains zero. The covariance is the cluster-robust one by economic region. Coastal adds the controls of column (2) of Table 7.

gives smaller effects than the instrumented one, 1.4 against 1.7 log points for income, 0.8 against 1.2 for employment and 5.2 against 6.4 for the price. Registered unemployment rises in 2015 and 2016 but does not survive the coastal controls, and the centrality classifications of Statistics Norway give the same results as the distance to the nearest large city. The 2015 change of source in the employment register is a level shift of about 3 percent that does not follow offshore work. Statistics Norway published the 2015 count of employees by workplace municipality both on the new source and adjusted to the old one, and the gap between the two is unrelated to exposure, in total, outside mining and manufacturing, and in public administration, education and health. On the adjusted series, jobs located in exposed municipalities fell by 0.8 per point of exposure from 2014 to 2015, against 1.0 unadjusted, so the fall in the year of the break is the shock and not the break.[8] The fall in prices did not come with more sales. Per point of offshore work, the number of free-market sales rose by 3 log points from 2013 to 2014 and by 2 from 2013 to 2017, neither of which can be told from zero, and sales per 1,000 dwellings moved by less than 1.5 in any year. A wave of forced sales would raise the number of transactions, and it does not appear, so prices fell on an unchanged volume.[9]

7 Conclusion

The oil price bust reached municipalities with almost no oil firms through the wages of their residents who worked on the shelf. Per percentage point of employed residents working offshore, income fell by about 2 percent and house prices by about 7 percent, while the dwelling stock and the population did not move. The effect works through the region, it is not the oil industry left in the sample,

[8] Table 16. The adjusted count carries 2014 forward with the change from the first quarter of 2015 to the first quarter of 2016, by industry group and workplace, so it is a bridge, not an observation, and it counts employees only.

[9] Figure 7. The sample is the 269 municipalities with at least 20 free-market sales in 2013, and the controls, weights and inference are those of Table 5. The count mixes forced and voluntary sales, and listings are not published by municipality.

and for each offshore job lost the equipment suppliers lost about one more and the rest of the local economy about two. Prices absorbed far more of the shock than income did, which is the question the paper leaves open.

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A Data sources

The supply-chain rule of 2003. Statistics Norway's list of petroleum-related industries (5-digit SN2007, table 13470) starts in 2008. Its correspondence table maps the list into SN2002 as follows: extraction and its support services (06, 09.1) into division 11; the building of ships, platforms and modules, installation work on them and ship repair (30.111, 30.113, 30.115, 30.116, 33.15) into division 35; test drilling (43.13) into 45.120; pipeline transport (49.5) into 60.300; offshore supply shipping (50.204) into 61.106; and supply bases (52.223) into 63.224. Services for the operation of pipelines (52.215, 136 jobs in 2008) have no entry. Of the list's 79,381 jobs located in municipalities in 2008, 90 percent fall into codes that map to divisions 11 and 35. Their share of a region's jobs in 2003 correlates 0.99 with the same share in 2008. Division 35 also contains small boats, leisure boats, rail and aircraft, so the 2003 rule is somewhat broader than the list.

Construction checks. Every analysis dataset is rebuilt from the raw downloads by the R scripts in the project, and a rebuild reproduces them exactly. The checks in `R/11_sanity_checks.R` confirm the following.

- Every national total reaches the 2019 map exactly, except that 22,000 jobs whose place of work is the shelf or unknown have no municipality in the place-of-work register.
- No municipal code with data is lost in the mapping.
- The event-study panel equals the long differences at every point where they measure the same change.

Table 11: Public data sources

Table	Content	Years used
Statistics Norway (StatBank), municipality level		
03311	Employed persons by broad SN2002 industry group (including division 11), by residence and by place of work	2003
03321	Employed persons by municipality of work and of residence (continental shelf)	2000–2019
08545	Employed persons by SN2002 two-digit industry, by place of work (shelf; supply chain, divisions 11 and 35), by residence (fishing, transport equipment, water transport) and by sex (shelf)	2000, 2003, 2008
13470	Employed persons by 5-digit SN2007 industry, place of work (petroleum-related list) and place of residence (industry of workplace, suppliers)	2008, 2013–2019
11365	Employees by place of work and 17 industry groups, 2015 adjusted for the change of source	2013–2015
11396	Employees by place of work and 17 industry groups, unadjusted	2015–2019
06944	Household income after tax and total, median, all households	2005–2019
11607	Employed persons 20–66 by immigrant background	2001–2019
11616	Employed residents 15–74	2000–2003
10540	Registered unemployed 15–74 (NAV), per cent	2000–2019
13563	Residents 15–74 by labour-market status	2008–2019
06913	Population on 1 January, births, deaths, growth	1970–2019
06265	Dwellings by building type, 1 January	2006–2019
05940	Dwellings started and completed	2000–2019
06035	Price per m^2 and number of sales, by dwelling type	2002–2019
06726	Free-market sales of dwellings, number and value	2000–2019
03068	Average income from employment and gross income per resident aged 17 and over, by sex (tax records)	2005–2019
12314	Employees, jobs and earnings by industry division (88 groups, SIC2007), national	2016
08050	Planned and actual petroleum investment (national)	2002–2019
04170	Accrued petroleum investment, older series (equal to 08050 where both exist)	2000–2019
14701	Consumer price index, annual	2000–2019
Statistics Norway, other		
Census	1980 Population and Housing Census, municipal booklets, table 9 (employed persons by place of work; row "North Sea")	1980
Statistics Norway, classifications (KLASS)		
131	Municipality code changes	1970–2019
108	Economic regions (2018 version, 2019 map)	2019
Corr. 108	Correspondence table SN2007–SN2002 (a different series of numbers from the classifications)	2009
Norwegian Offshore Directorate (FactPages and FactMaps)		
Fields	Outlines of the 139 oil and gas fields	current
Other		
Eurostat GISCO	County boundaries (NUTS 2016 level 3)	2016
FRED (EIA)	Brent spot price, daily (DCOILBRENTEU)	2000–2019
Wikidata	Municipal administrative centres	current
Heliports	Coordinates of the five offshore heliports in use in 2003	2003

B Employment by industry

Table 12: Employed residents by industry of workplace, change per 100 employed residents

	2013–17 (1)	2013–19 (2)	2015–19 (3)
All industries	−1.44***	−1.49**	−0.60*
	(0.38)	(0.52)	(0.33)
Oil and gas, mining and quarrying (05–09)	−0.32***	−0.20***	−0.09
	(0.06)	(0.07)	(0.06)
All other industries	−1.11***	−1.29**	−0.51
	(0.36)	(0.51)	(0.33)
Agriculture, forestry and fishing (01–03)	0.21***	0.23**	0.11
	(0.07)	(0.09)	(0.07)
Manufacturing and mining outside oil (10–33)	−0.53**	−0.59**	−0.40**
	(0.22)	(0.23)	(0.14)
Electricity, water and waste (35–39)	0.00	−0.02	−0.03
	(0.02)	(0.03)	(0.02)
Construction (41–43)	−0.08	−0.11	−0.08
	(0.09)	(0.12)	(0.09)
Trade (45–47)	−0.13	−0.19*	−0.03
	(0.09)	(0.10)	(0.09)
Transport and storage (49–53)	−0.13***	−0.17***	−0.01
	(0.04)	(0.05)	(0.04)
Hotels and restaurants (55–56)	−0.25***	−0.24***	−0.05
	(0.05)	(0.08)	(0.06)
Information and communication (58–63)	−0.02	−0.06	−0.07
	(0.04)	(0.06)	(0.05)
Finance and insurance (64–66)	−0.04	−0.05*	−0.02
	(0.02)	(0.03)	(0.02)
Real estate, professional and technical services (68–75)	−0.15**	−0.17*	−0.11
	(0.05)	(0.08)	(0.07)
Business services (77–82)	−0.05	−0.03	0.08*
	(0.06)	(0.08)	(0.05)
Public administration (84)	0.08	0.05	−0.03
	(0.06)	(0.06)	(0.06)
Education (85)	0.06	0.09	0.04
	(0.06)	(0.10)	(0.08)
Health and social work (86–88)	−0.06	0.05	0.08
	(0.13)	(0.17)	(0.13)
Other services (90–99)	0.00	−0.05	−0.01
	(0.04)	(0.05)	(0.04)
Unspecified	−0.01	−0.02	0.01
	(0.02)	(0.02)	(0.03)

Note: Coefficients on offshore work in 2003, in percentage points of employed residents, in the 375 municipalities of Table 5. The outcome is the change in the number of employed residents whose workplace is in the industry, fourth quarter of the first year to fourth quarter of the last, per 100 employed residents in 2013 (Statistics Norway table 13470, by place of residence, SIC 2007 groups in parentheses). The register changed source in 2015, so columns (1) and (2) contain the break and column (3) lies inside the new source. Controls, weights, clustering and stars as in Table 5.

Table 13: Employed residents in the petroleum-related industries and in manufacturing, change per 100 employed residents

	2013–17 (1)	2013–19 (2)	2015–19 (3)
Petroleum-related industries, Statistics Norway's list	-0.40^{***}	-0.33^{**}	-0.12
	(0.10)	(0.11)	(0.06)
Extraction and support services (06, 09.1)	-0.35^{***}	-0.22^{***}	-0.11^{*}
	(0.05)	(0.07)	(0.06)
Platforms, ships and ship repair (30.11, 33.15)	0.00	-0.04	0.01
	(0.09)	(0.09)	(0.06)
Pipelines, supply vessels and terminals	-0.05^{***}	-0.06^{**}	-0.02
	(0.02)	(0.03)	(0.02)
Manufacturing outside the list	-0.53^{**}	-0.55^{**}	-0.41^{**}
	(0.20)	(0.22)	(0.14)
Machinery and equipment (28)	-0.28^{***}	-0.28^{**}	-0.25^{**}
	(0.11)	(0.11)	(0.11)
Fabricated metal products (25)	-0.12^{**}	-0.15^{***}	-0.08^{*}
	(0.04)	(0.05)	(0.04)
Computer, electronic and optical products (26)	-0.10	-0.09	-0.05
	(0.08)	(0.07)	(0.04)
Chemicals (20)	-0.02	-0.07^{*}	-0.05^{***}
	(0.04)	(0.04)	(0.02)
Ships, boats and other transport equipment (30)	-0.07	-0.10	-0.02
	(0.08)	(0.09)	(0.06)

Note: As Table 12. The petroleum-related industries are Statistics Norway's list (five-digit codes 06, 09.1, 30.111, 30.113, 30.115, 30.116, 33.15, 43.13, 49.5, 50.204, 52.215 and 52.223); manufacturing outside the list is divisions 10–33 less the list's manufacturing codes.

Table 14: Jobs lost for each offshore job lost, with the suppliers purged

	2013–17 (1)	2013–19 (2)	2015–19 (3)
Panel A. Change in employed residents, per 100 employed residents in 2013			
Offshore jobs: extraction and support services (06, 09.1)	-0.35^{***}	-0.22^{***}	-0.11^{*}
	(0.05)	(0.07)	(0.06)
Suppliers: the list's other codes, divisions 25 and 28	-0.45^{***}	-0.53^{***}	-0.34^{***}
	(0.14)	(0.13)	(0.12)
Both, the direct hit	-0.79^{***}	-0.76^{***}	-0.45^{***}
	(0.16)	(0.16)	(0.12)
The rest of the local economy	-0.64^{*}	-0.74	-0.16
	(0.33)	(0.48)	(0.34)
also without the other manufacturing divisions	-0.51	-0.62	-0.08
	(0.29)	(0.46)	(0.34)
All but mining and quarrying (05–09)	-1.11^{***}	-1.29^{**}	-0.51
	(0.36)	(0.51)	(0.33)
All industries	-1.44^{***}	-1.49^{**}	-0.60^{*}
	(0.38)	(0.52)	(0.33)
Panel B. Jobs lost for each offshore job lost			
The rest of the local economy	1.85	3.29	1.50
	(0.92)	(2.30)	(3.39)
	[−0.1, 3.8]	[−1.3, 9.7]	[−14.5, 41.8]
also without the other manufacturing divisions	1.46	2.74	0.75
	(0.87)	(2.23)	(3.32)
	[−0.2, 3.6]	[−1.5, 9.4]	[−20.7, 31.3]
All but mining and quarrying (05–09)	3.21	5.77	4.89
	(0.99)	(2.63)	(4.50)
	[1.3, 5.5]	[1.5, 14.6]	[−17.8, 75.5]

Note: Panel A as Table 12. The suppliers are the petroleum list's codes other than extraction and support services plus fabricated metal products (25) and machinery and equipment (28), the firms described in Section 5.3, which the sample rule does not remove. The rest of the local economy is all employed residents less the direct hit, and the next row also removes the other manufacturing divisions. In 2013 the offshore jobs were 0.7 and the supplier jobs 1.1 per 100 employed residents of the sample. Panel B divides each remainder's coefficient by the offshore coefficient of the same column, with delta-method standard errors from a stacked regression with region clusters in parentheses and 95 percent percentile intervals from a pairs cluster bootstrap over regions (999 draws) in brackets. Over 2015–19 the offshore loss itself is small and imprecise, so the ratios of column (3) are not informative.

Table 15: Jobs lost for each offshore job lost on the supply-chain samples, 2013–17

	Headline (1)	Supply chain ≤ 2% (2)	Supply chain ≤ 1% (3)
Panel A. Change in employed residents 2013–17, per 100 employed residents in 2013			
Offshore jobs: extraction and support services (06, 09.1)	−0.35***	−0.28***	−0.34***
	(0.05)	(0.05)	(0.03)
Suppliers: the list's other codes, divisions 25 and 28	−0.45***	−0.50***	−0.64***
	(0.14)	(0.21)	(0.31)
Both, the direct hit	−0.79***	−0.78***	−0.98***
	(0.16)	(0.21)	(0.30)
The rest of the local economy	−0.64*	−0.91	−0.52
	(0.33)	(0.47)	(0.54)
also without the other manufacturing divisions	−0.51	−0.97*	−0.56
	(0.29)	(0.40)	(0.40)
All but mining and quarrying (05–09)	−1.11***	−1.42**	−1.18*
	(0.36)	(0.48)	(0.59)
All industries	−1.44***	−1.69***	−1.50**
	(0.38)	(0.47)	(0.58)
Panel B. Jobs lost for each offshore job lost			
The rest of the local economy	1.85	3.26	1.53
	(0.92)	(1.96)	(1.60)
	[−0.1, 3.8]	[−0.7, 7.5]	[−1.4, 6.0]
also without the other manufacturing divisions	1.46	3.47	1.65
	(0.87)	(1.78)	(1.19)
	[−0.2, 3.6]	[0.1, 7.7]	[−0.6, 5.6]
All but mining and quarrying (05–09)	3.21	5.08	3.45
	(0.99)	(2.21)	(1.84)
	[1.3, 5.5]	[0.7, 9.9]	[−0.2, 8.5]
Municipalities	375	290	250

Note: As Table 14, window 2013–17. Column (1) is the headline sample. Columns (2) and (3) keep the headline municipalities whose economic region had at most 2 and at most 1 percent of its 2003 jobs in SN2002 divisions 11 and 35, extraction and the building of ships and platforms, the supply-chain rule of Section 3.2 and Table 3. Table 3 applies the rule to all municipalities; here it is applied inside the headline sample, which is why the counts are 290 and 250 rather than 296 and 256.

C The 2015 change of source in the employment register

Table 16: The 2015 register break by municipality and offshore work

	Mean	Offshore work, 2003	(SE)	Municipalities
Break, all industries	−3.57	−0.23	(0.33)	375
Break, outside mining and manufacturing	−3.74	−0.24	(0.31)	375
Break, public administration, education and health	−1.52	−0.10	(0.38)	375
Change 2014–15, adjusted	0.75	−0.80***	(0.28)	375
Change 2014–15, unadjusted	−2.83	−1.03***	(0.29)	375
Change 2013–14	1.32	0.06	(0.25)	375
Change 2013–17, bridged across the break	5.00	−1.17	(0.67)	375
Change 2013–17, unadjusted	1.42	−1.40**	(0.59)	375
Change 2015–17	2.93	−0.43	(0.38)	375

Note: Employees aged 15–74 by municipality of workplace in the fourth quarter, mapped to the 2019 municipalities, from the two 2015 counts described in Section 6.3: the unadjusted count on the new source and the count adjusted to be comparable with 2014. The break is 100 times the log difference between them. The changes are 100 times log differences, and the bridged 2013–17 change joins 2013 to the adjusted 2015 and the unadjusted 2015 to 2017. Mean is the population-weighted mean in the 375 municipalities of Table 5; the coefficient is on offshore work in 2003, with the controls, weights, inference and stars of Table 5.

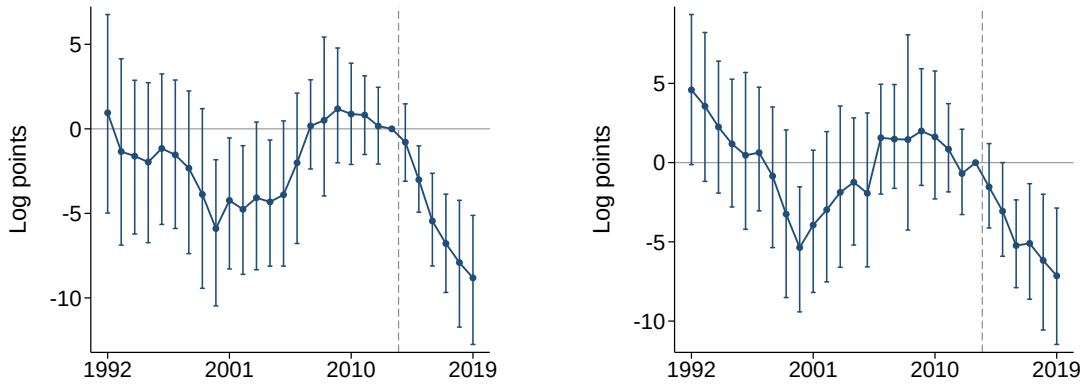
D Pre-boom placebos

Table 17: Pre-boom outcomes on the 2003 offshore share, by sample

Sample	Population growth		Employment	Registered	Municipalities
	1980–90	1990–2003	growth 2000–03	unemployment 2000–03	
	(1)	(2)	(3)	(4)	
Headline sample	0.93	0.38	0.42	0.02	375
	(0.79)	(0.75)	(0.38)	(0.08)	
Balanced price panel	0.75	0.27	0.31	0.06	271
	(0.82)	(0.79)	(0.38)	(0.08)	
Supply chain at most 2 percent	1.23	1.08	0.63	0.09	290
	(1.11)	(1.21)	(0.61)	(0.13)	
Supply chain at most 1 percent	1.23	2.27	1.28**	−0.02	250
	(1.36)	(1.34)	(0.59)	(0.14)	
Without hub regions, 5 percent	1.24	1.01	0.81	−0.05	272
	(1.18)	(1.06)	(0.52)	(0.12)	
Without hub regions, 3 percent	0.18	1.40	1.09	−0.02	240
	(1.39)	(1.48)	(0.59)	(0.16)	

Note: Coefficients on offshore work in 2003, in percentage points of employed residents, with the regional onshore oil share, log population in 2004 and the 2003 industry shares as controls, population weights of 2004, standard errors clustered by economic region in parentheses and stars from the region wild bootstrap, * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Population growth is 100 times the log change, employment growth 100 times the log change in employed residents from 2000 to 2003, and registered unemployment the change in percentage points. The samples are those of Table 5 (headline), the balanced price panel of Table 5, column (3), the two supply-chain samples of Table 15 and the headline sample without the hub regions of Table 7, columns (4) and (5). Municipalities are those of column (3); columns (1), (2) and (4) have one fewer.

E Further checks



(a) Baseline controls **(b)** Coastal controls

Figure 5: The average sale price from 1992, relative to 2013

Note: Coefficients on offshore work in 2003 by year, 100 times the log difference of the average sale price from 2013, with 95 percent region wild bootstrap intervals, on the municipalities of the headline sample with a price in every year from 1992 to 2019 (Statistics Norway table 06726). Panel (b) adds the coastal controls of Section 3.4. A joint test that the coefficients of 1992 to 2002, with base year 2003, are zero has a p-value of 0.30 in panel (a) and 0.05 in panel (b).

Table 18: The crisis cushion of 2008 and 2009

	Baseline (1)	Coastal controls (2)
Employment income per adult man	0.63^{***}	0.38^{***}
	(0.14)	(0.12)
Employment income per adult woman	0.26^{*}	0.21
	(0.13)	(0.13)
Median household income after tax	0.24	0.16
	(0.13)	(0.11)
Registered unemployment rate	−0.01	0.02
	(0.04)	(0.04)
Average sale price	0.88	0.34
	(1.53)	(1.51)
Detached-house price per m^2	1.87^{*}	1.16
	(0.82)	(1.02)
Apartment price per m^2	4.47^{***}	5.22^{***}
	(1.02)	(1.45)
Number of free-market sales	0.71	2.02
	(1.33)	(1.80)

Note: Each entry is $(\beta_{2008} + \beta_{2009})/2 - (\beta_{2006} + \beta_{2007} + \beta_{2010} + \beta_{2011})/4$, the average event-study coefficient of 2008 and 2009 relative to the two years on either side, per percentage point of offshore work in 2003, from event studies with base year 2013 on the headline sample from 2005, as in Figure 3. Incomes and prices are in log points, registered unemployment in percentage points. Column (2) adds the coastal controls of Section 3.4. Standard errors clustered by economic region in parentheses and stars from the region wild bootstrap as in Table 5. The detached and apartment prices per square metre are observed where Statistics Norway publishes them.

Table 19: The 2003 share, the 2013 share and the instrumented 2013 share, bust 2013–2017

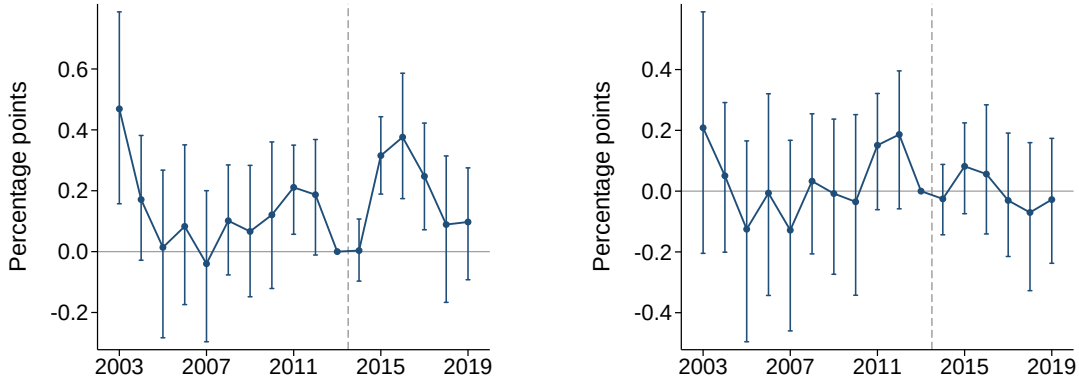
	Income (1)	Employment (2)	Price (3)
Offshore work, 2003	-1.932^{***}	-1.378^{***}	-7.142^{***}
	(0.385)	(0.386)	(1.444)
Offshore work, 2013, least squares	-1.385^{***}	-0.810^{***}	-5.174^{***}
	(0.259)	(0.290)	(1.244)
Offshore work, 2013, instrumented	-1.712^{***}	-1.221^{***}	-6.400^{***}
	(0.374)	(0.388)	(1.435)
Municipalities	375	375	271

Note: Coefficients per percentage point of offshore work, with the controls, weights and inference of Table 5. The first row is panel A of Table 5, the second regresses the same outcomes on the 2013 share by least squares, and the third is panel B of Table 5, the 2013 share instrumented by the 2003 share.

Table 20: The bust with the centrality classifications of Statistics Norway, 2013–2017

	Income (1)	Employment (2)	Price (3)
Baseline	−1.932***	−1.378***	−7.142***
	(0.385)	(0.386)	(1.444)
Distance to the nearest large city	−1.824***	−1.106***	−6.219***
	(0.328)	(0.345)	(1.450)
Centrality classes, 2008	−2.008***	−1.213**	−7.099***
	(0.383)	(0.454)	(1.532)
Centrality groups, 2018	−1.962***	−1.062**	−6.340***
	(0.327)	(0.394)	(1.584)
Municipalities	375	375	271

Note: Coefficients on offshore work in 2003 with the specification, weights and inference of Table 5. The second row adds the distance to the nearest of Oslo, Bergen, Trondheim and Stavanger, as in Table 8, the third the 2008 centrality standard of Statistics Norway as three class dummies, and the fourth its 2018 centrality index as five group dummies.

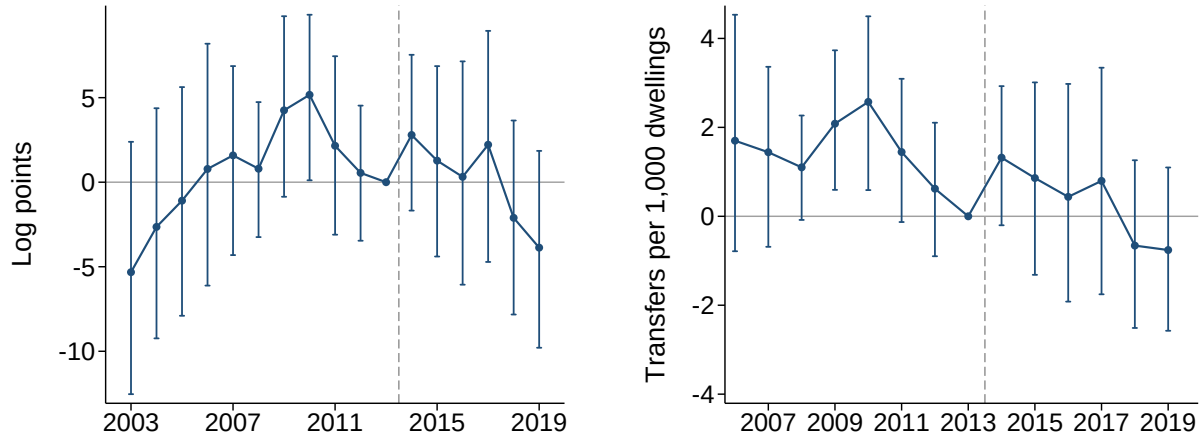


(a) Baseline controls **(b)** Coastal controls

Figure 6: Registered unemployment, 2003–2019

Note: Coefficients on offshore work in 2003 by year, the change in the registered unemployment rate from November 2013 in percentage points, with 95 percent region wild bootstrap intervals, on the headline sample. Panel (b) adds the coastal controls of Section 3.4.

F Sales volumes



(a) Number of free-market sales, log points

(b) Sales per 1,000 dwellings

Figure 7: Offshore work and the number of dwelling sales, 2003–2019

Note: As Figure 3, in calendar years, for the number of free-market sales, 100 times the log difference from 2013, and for sales per 1,000 dwellings minus their 2013 value, which starts in 2006 with the dwelling stock series, in the 269 sample municipalities with at least 20 free-market sales in 2013.